Python For Loop – Important Notes

Complete beginner-friendly notes with syntax, rules, range(), examples, nested loops, control statements, and common programs.

1. What is a Loop?

A loop is used to execute a block of code repeatedly. Python mainly provides **for loop** and **while loop**. A for loop is commonly used when we want to iterate over a sequence or a collection.

2. What is a For Loop?

A **for loop** is used to repeat a block of statements for each item in an iterable such as a list, tuple, string, set, dictionary, or range.

3. Basic Syntax

```
for variable in sequence:
    statement
```

The loop variable receives one item at a time from the sequence.

4. Simple Example

```
for i in range(1, 6):
    print(i)
```

Output: 1 2 3 4 5

5. Important Points

• Python uses indentation to define the loop body.

• The **for** keyword starts a for loop.

• The loop variable changes on every iteration.

• A for loop can work with any iterable.

• The **range()** function is frequently used with for loops.

• A loop can contain conditions and other loops.

6. for Loop with String

```
name = "Python"

for ch in name:
    print(ch)
```

7. for Loop with List

```
numbers = [10, 20, 30, 40]

for num in numbers:
    print(num)
```

8. range() Function

range() generates a sequence of numbers. It is very commonly used in for loops.

Form	Meaning	Example
range(stop)	Starts from 0 and stops before stop	range(5) → 0,1,2,3,4
range(start, stop)	Starts at start and stops before stop	range(2,6) → 2,3,4,5
range(start, stop, step)	Moves by step	range(1,10,2) → 1,3,5,7,9

9. range() Examples

Print 1 to 10

```
for i in range(1, 11):
    print(i)
```

Print even numbers

```
for i in range(2, 21, 2):
    print(i)
```

Print odd numbers

```
for i in range(1, 20, 2):
    print(i)
```

Reverse counting

```
for i in range(10, 0, -1):
    print(i)
```

10. for Loop with if-else

```
for i in range(1, 11):
    if i % 2 == 0:
        print(i, "Even")
    else:
        print(i, "Odd")
```

11. break Statement

break immediately stops the loop.

```
for i in range(1, 11):
    if i == 5:
        break
    print(i)
```

12. continue Statement

continue skips the current iteration and moves to the next iteration.

```
for i in range(1, 6):
    if i == 3:
        continue
    print(i)
```

13. pass Statement

pass does nothing. It is used as a placeholder when a statement is required.

```
for i in range(5):
    pass
```

14. Nested for Loop

A loop inside another loop is called a nested loop.

```
for i in range(1, 4):
    for j in range(1, 4):
        print(i, j)
```

15. for Loop with else

The else block executes when the loop finishes normally. If the loop is stopped using break, the else block is skipped.

```
for i in range(1, 4):
    print(i)
else:
    print("Loop completed")
```

16. Common Programs

Sum of numbers

```
total = 0
for i in range(1, 6):
    total = total + i
print(total)
```

Count even numbers

```
count = 0
for i in range(1, 11):
    if i % 2 == 0:
        count = count + 1
print(count)
```

Multiplication table

```
num = 5
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

Factorial

```
num = 5
fact = 1
for i in range(1, num + 1):
    fact = fact * i
print(fact)
```

Find largest number

```
numbers = [10, 25, 7, 40, 15]
largest = numbers[0]
for num in numbers:
    if num > largest:
        largest = num
print(largest)
```

17. for Loop vs while Loop

for loop	while loop
Used to iterate over a sequence/iterable	Runs while a condition is true
Often used when number of iterations is known	Often used when iterations depend on a condition
Example: for i in range(5)	Example: while i < 5

18. Important Interview Points

1. What is a for loop?

2. What is an iterable?

3. How does range() work?

4. Difference between range(5) and range(1, 5).

5. What is the use of break?

6. What is the use of continue?

7. What is a nested loop?

8. When does the else block of a for loop execute?

9. Can a for loop iterate over a string? Yes.

10. Can a for loop iterate over a dictionary? Yes.

19. Quick Revision

for → repeats for each item
range() → generates numbers
break → stops the loop
continue → skips current iteration
pass → does nothing / placeholder
nested loop → loop inside another loop